

BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

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Abstract

In same-domain legal consultation archives, dense retrieval often surfaces a relevant client question while failing to return the lawyer’s advice turn—an *episode incompleteness* failure that leaves prior recommendations hard to recover under lexical interference from neighbouring matters. BIMS-LEGAL indexes Chinese consultation logs in a dual store: a fast episodic turn index with session identifiers and a slower semantic store consolidated by target- k KMeans. The design target is episode completeness under same-domain interference: Soft O2 binds session siblings by attenuated score inheritance (βs) rather than hard score copy, so a retrieved question can surface its advice turn without dumping an entire session into top- k . Exact replay is a surface-form reference; main channels are paraphrase, follow-up, and advice-recall. On a shared turn-level FlatIP rebuild, Soft O2 raises Answer Hit over dense FlatIP; hard hydration, which copies sibling scores without attenuation, can raise AH further by forcing siblings into top- k . Soft O2-C is evaluated as a conditional cluster-binding mechanism under Mix reconstructions with off-session gold; gated Hybrid is the composed operator when thematic co-membership helps. We report Answer Hit, Episode Completeness, graded nDCG with fixed gold-session IDCG, a four-way failure taxonomy, and Holm-adjusted Soft O2 vs FlatIP tests whose ΔAH matches the Soft O2 grids by construction.

Keywords: legal information retrieval, consultation memory, soft episode binding, dual-store memory, cluster binding, Chinese legal corpora

1. Introduction

2 A junior associate may ask a firm assistant: “Last month you advised
3 this client on terminating a fixed-term contract—what remedy did we rec-
4 ommend?” The system returns several near-identical “contract termination”

5 questions, ranks the stored user utterance highly, and never surfaces the part-
 6 ner’s written advice. The subsequent reply may sound fluent yet be grounded
 7 in the wrong episode—or in none. In consultation archives, questions and
 8 answers share specialized terminology, neighbouring matters are topically ad-
 9 jacent yet legally distinct, and the professionally useful evidence unit is the
 10 full consultation *episode*.

11 Legal informatics has long studied statutory interpretation, case retrieval,
 12 and advisory systems Martinez-Gil (2023); Ashley (2017), and recent legal
 13 large language models (LLMs) further improve statutory and fact-pattern
 14 answering Yue et al. (2023); Huang et al. (2023); Yue et al. (2024). Classical
 15 legal information retrieval (IR), however, mainly targets *external* authority—
 16 statutes, cases, or community answers—as documents Askari et al. (2024);
 17 Louis et al. (2024); Liu et al. (2022); Zhang et al. (2024). Memory-augmented
 18 dialogue systems recover evidence from evolving conversation logs Wu et al.
 19 (2024); Maharana et al. (2024); Packer et al. (2023), yet firm assistants still
 20 need a reliable index of *internal consultation memory*: prior client-specific
 21 advice buried in a growing same-domain archive under lexical interference
 22 from neighbouring matters.

23 Under that interference, dense turn retrieval often returns a question
 24 turn while its answer sibling remains outside top- k . We term this failure
 25 *episode incompleteness*, as distinct from a full session miss. It is worsened
 26 when approximate indexes distort turn scores and fragment episode context
 27 at moderate archive sizes. Parent-document hydration and session-max ag-
 28 gregation can force siblings into the result list by hard score copy, but they
 29 also erase rank competition among unrelated candidates. What is needed is
 30 a way to complete episodes without collapsing the retrieval ranking into a
 31 session dump.

32 This paper presents BIMS-LEGAL, a dual-store system for Chinese legal
 33 consultation logs. A fast episodic pathway indexes timestamped turn embed-
 34 dings with speaker role and session identifier *sid*; a slower semantic pathway
 35 consolidates turns with target- k KMeans. The dual-pathway split is inspired
 36 by Complementary Learning Systems theory McClelland et al. (1995); Ku-
 37 maran et al. (2016) only as an architectural analogy. On the episodic pathway,
 38 Soft O2 performs session soft binding: when a turn is retrieved with score s ,
 39 siblings sharing *sid* inherit $\max(s_{\text{t}}, \beta \cdot s)$, so an answer turn can surface under
 40 question-hit / answer-miss while attenuated scores keep candidate competi-
 41 tion intact. On the semantic pathway, Soft O2-C applies the same rule within
 42 KMeans clusters when gold evidence is not co-session with the query; gated

Hybrid is intended to trigger cluster binding only on direct dense hits. Exact FlatIP indexing (O1) restores faithful turn scores when product quantization collapses Answer Hit; bulk-load consolidation (O3) makes large shared-store construction tractable without changing retrieval semantics. We evaluate Soft O2 primarily against FlatIP under episode incompleteness, treat hard hydration as an unrestricted score-copy upper bound, and examine Soft O2-C under Mix reconstructions as a conditional mechanism test.

The main contributions are as follows. First, we formulate consultation-memory retrieval under same-domain interference and identify episode incompleteness as a recurrent failure mode of dense turn retrieval. Second, we propose Soft O2 as a turn-level soft-binding operator: a gap-ratio score-competition analysis identifies a feasible high- β band, and an Answer Hit grid search over $\{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$ selects default $\beta=0.98$ (Table A.4), with O1/O3 as implementation operators for shared-store evaluation. Third, we evaluate Soft O2 on CAIL multi-turn and rebuilt LegalEp corpora against FlatIP, hard hydration, shuffled-*sid*, BM25-turn, RRF, and cross-encoder controls under a unified FlatIP-rebuild metric pipeline, and report Soft O2-C Mix contrasts as conditional mechanism evidence with graded nDCG, a four-way failure taxonomy, and Holm-adjusted Soft O2 vs FlatIP tests.

Section 2 reviews related work. Section 3 presents BIMS-LEGAL and the Soft O2 attenuation analysis. Section 4 details corpora, metrics, and protocols. Section 5 reports results. Section 6 discusses implications and limitations, and Section 7 concludes.

2. Related Work

2.1. Legal IR and long-horizon dialogue memory

Legal QA and retrieval systems typically ground generation in statutes, cases, or published advice Louis et al. (2024); Askari et al. (2024); Martinez-Gil (2023); Ashley (2017). Conversational legal case retrieval and event-aware similar-case ranking further show how interaction context and structured legal knowledge shape IR effectiveness Liu et al. (2022); Zhang et al. (2024). Those pipelines optimize access to external authority. Consultation memory retrieval targets a different evidence source: *prior consultation episodes* inside an agent’s own archive. Confusing two clients’ histories can produce fluent yet professionally inappropriate responses even when statute retrieval is correct. Domain LLMs such as DISC-LawLLM, Lawyer LLaMA, and LawLLM improve statutory and advisory generation Yue et al. (2023); Huang et al.

(2023); Yue et al. (2024), but they still need a reliable memory index when prior advice must be recovered from a growing same-domain log.

Legal consultation archives add a further difficulty: high lexical overlap among topically adjacent matters, with the professionally useful evidence unit being the full question-answer episode. This setting motivates *dual-store* memory architectures that keep fast episodic turn retrieval separate from slower semantic clustering, and session-aware soft binding operators that recover answer turns under same-domain interference without collapsing rank competition among unrelated candidates.

2.2. Dual-store memory and episodic retrieval

Dual-store designs distinguish fast episodic encoding from slower semantic integration McClelland et al. (1995); Kumaran et al. (2016); Tulving et al. (1972); Atkinson and Shiffrin (1971). Memory architectures for dialogue agents extend context via hierarchical paging, summary banks, graphs, and neuro-symbolic retrieval Packer et al. (2023); Zhong et al. (2024); Rasmussen et al. (2025); Gutiérrez et al. (2024); Modarressi et al. (2023); Wang et al. (2024); Lewis et al. (2020); Wu et al. (2025), but most treat retrieved units as flat passages rather than *multi-turn consultation episodes* whose answer turn may rank below an unrelated question turn sharing legal vocabulary. LongMemEval and LoCoMo evaluate long-horizon recall and QA over personal conversation logs Wu et al. (2024); Maharana et al. (2024), while broader memory-agent surveys emphasize persistence beyond the context window Wu et al. (2025); Tan et al. (2025). Recent conversational RAG work stresses reuse of historical search evidence across turns Mo et al. (2026). BIMS-LEGAL positions its dual-store contribution at this intersection: episodic turn embeddings with *sid* for same-session recovery (Soft O2), and KMeans semantic centroids with Soft O2-C for cross-session recovery when query and gold evidence occupy different sessions but share thematic structure. Under dense same-domain interference, a question turn can occupy top- k while its answer sibling is absent. Parent-document hydration and session-max aggregation recover that sibling by hard score copy; Soft O2 instead injects attenuated inherited scores so episode completion does not erase rank competition among unrelated candidates.

2.3. Session-aware retrieval

Parent-document hydration, session aggregation, and joint session indexing are standard session-aware mechanisms for attaching sibling context to a

115 dense hit. Soft episode binding (Soft O2) inherits $\beta \cdot s$ for siblings and keeps
 116 ranking competition among candidates. Parent-document and hierarchical
 117 retrieval similarly attach child evidence to a parent unit Callan et al. (1994);
 118 Dai et al. (2019); Zaheer et al. (2017); Karpukhin et al. (2020); Soft O2
 119 differs by using attenuated turn-level score propagation rather than hard hy-
 120 dration or atomic session documents. Lexical BM25 Robertson and Zaragoza
 121 (2009), dense passage retrieval Karpukhin et al. (2020), dense-sparse Recipro-
 122 cal Rank Fusion Cormack et al. (2009), and cross-encoder reranking Nogueira
 123 et al. (2019); Chen et al. (2024) provide complementary controls outside ses-
 124 sion membership. Exact flat indexes and product-quantized ANN backends
 125 Johnson et al. (2019) further affect whether sibling binding is applied over
 126 faithful turn scores.

127 *Contrast with joint-episode indexing..* An alternative way to keep episode
 128 context is to concatenate full question-answer pairs into monolithic episode
 129 documents (joint indexing). Joint indexing can prevent episode incomplete-
 130 ness by construction, but it changes the evidence unit from individual turns to
 131 merged documents. That coarser unit weakens fine-grained turn matching—
 132 for example, isolating a specific advice sentence—and raises index memory
 133 and update cost when new turns arrive in an online consultation log. Soft O2
 134 instead keeps turn-level representations and candidate competition, recover-
 135 ing missing siblings on demand via attenuated score inheritance (βs) with-
 136 out replacing the underlying turn index. Table 1 summarizes the session-
 137 structure contrasts used in the main grids.

Table 1: Session-aware mechanisms compared in this work.

Mechanism	Unit	Expansion	Type	Rank
Parent hydration	Session	Insert siblings	Hard	No
Session-max	Session	Session max	Hard	Same*
Joint episode index	Session doc	Atomic	N/A	N/A
Soft O2	Turn+ <i>sid</i>	Soft βs	Soft	Yes
Soft O2-C	Turn+cluster	Soft $\beta_c s$	Soft	Yes

138 *On the reported corpora, session-max rankings coincide with hard parent hydration
 139 under full-sibling expansion; main tables report a single hard-expansion baseline. Soft O2
 140 denotes session soft binding; Soft O2-C denotes cluster soft binding on the KMeans se-
 141 mantic pathway.

142 In short, the dual-store split is the system design, and Soft O2 /Soft O2-C
 143 are the operators evaluated on Chinese legal consultation archives.

144 3. Method: BIMS-LEGAL

145 3.1. Architecture overview

146 Figure 1 summarizes the pipeline. BIMS-LEGAL maintains complemen-
 147 tary episodic and semantic stores over the same consultation archive.

148 **Definition 1** (Episodic and semantic elements). *An episodic turn (episodic
 149 element) is $e_i = (\mathbf{v}_i, t_i, c_i, \phi_i, \text{sid}_i)$, where $\mathbf{v}_i \in \mathbb{R}^d$ is an embedding, t_i a times-
 150 tamp, c_i conversational context (including speaker role), ϕ_i surface features,
 151 and sid_i the consultation session identifier. We reserve session turns for the
 152 weaker notion of turns that share only sid under a grouping key, without re-
 153 quiring role, timestamp, or other episodic metadata to participate in scoring.
 154 Soft O2 binds over session siblings $\text{Sib}(t) = \{t' : \text{sid}_{t'} = \text{sid}_t\} \setminus \{t\}$, but
 155 the indexed objects themselves remain episodic turns. A semantic element is
 156 $s_j = (\mathbf{c}_j, C_j, \kappa_j, \tau_j)$ with centroid $\mathbf{c}_j$, member set C_j , label κ_j , and formation
 157 statistics τ_j . Write $\text{Clus}(t) = \{t' : \text{cid}_{t'} = \text{cid}_t\} \setminus \{t\}$ for semantic-cluster
 158 siblings.*

159 **Definition 2** (Dual-store retrieval map). *Given query q with embedding
 160 $\mathbf{v}_q$, base episodic retrieval returns a scored candidate set $\mathcal{R}_0(q) = \{(t, s_t)\}$
 161 under Eq. (1). Soft O2 produces $\mathcal{R}_{\text{O2}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Sib}, \beta)$ (Sec-
 162 tion 3.3). Soft O2-C produces $\mathcal{R}_{\text{O2C}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Clus}, \beta_c)$ (Sec-
 163 tion 3.5). Gated Hybrid composes session binding then cross-session cluster
 164 binding (Algorithm 2). The returned list is the top- k of the bound set under
 165 the rewritten scores.*

166 3.2. Base scoring and implementation operators (O1, O3)

167 For query q with embedding $\mathbf{v}_q$, episodic base retrieval scores a memory
 168 turn m by

$$\text{score}(m) = \alpha S(\mathbf{v}_q, \mathbf{v}_m) + (1 - \alpha) R(t_m), \quad (1)$$

169 where S is cosine similarity on ℓ_2 -normalized embeddings and

$$R(t_m) = \max\left(0, 1 - \frac{\Delta_t}{30 \text{ days}}\right), \quad (2)$$

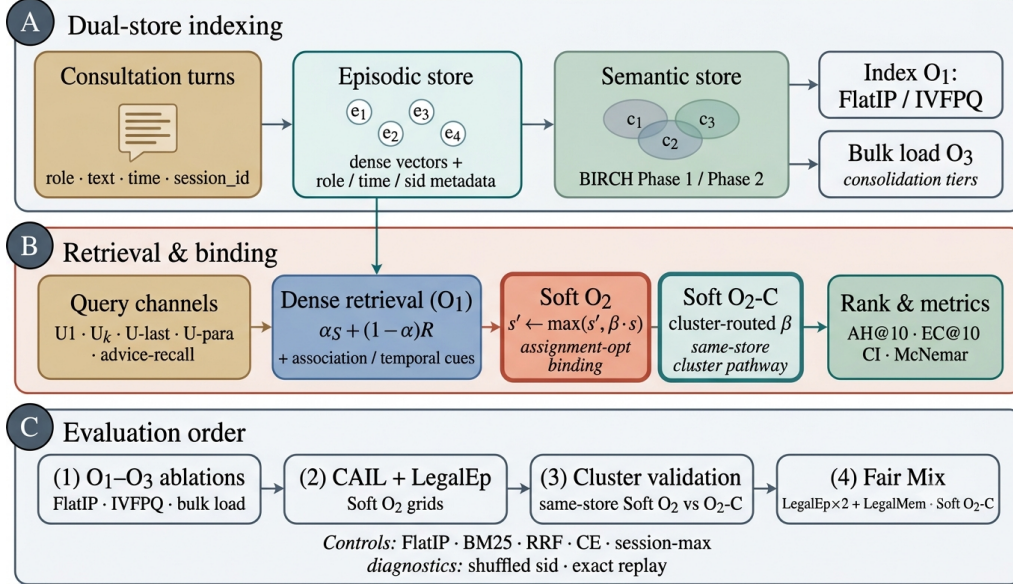


Figure 1: BIMS-LEGAL dual-store pipeline. (A) Episodic turn indexing and KMeans semantic consolidation (O3 bulk-load). (B) Query-time FlatIP retrieval with Soft O2 session binding and Soft O2-C cluster binding; O1 and O3 are implementation operators. (C) Evaluation order: shared-store ablations, Soft O2 grids, and Mix Soft O2-C.

170 with Δ_t the elapsed time between query time and the turn timestamp. Unless
 171 stated otherwise, $\alpha=0.7$. Eq. (1) is the ranking score used throughout the
 172 legal grids; Soft O2 /Soft O2-C rewrite sibling scores after this fusion step
 173 and do not replace Eq. (1).

174 *O1: Exact FlatIP indexing..* Let $\hat{S}$ denote the similarity induced by an ANN
 175 backend. At hundreds to a few thousand turns, FAISS IVFPQ is often under-
 176 trained, so $\|\hat{S} - S\|$ is large enough to reorder turns and break Soft O2
 177 inheritance. O1 forces $\hat{S} = S$ via exact FlatIP (inner-product search on
 178 ℓ_2 -normalized embeddings). Quantization error depresses Answer Hit (Ta-
 179 ble A.3: IVFPQ AH 0.330 at $M=1600$ on LegalEp-DISC exact). O1 is an
 180 engineering prerequisite for Soft O2 evaluation, not a standalone retrieval
 181 algorithm.

182 *O3: Bulk-load consolidation..* During bulk construction, O3 replaces the per-
 183 insert update ($e_i \mapsto \text{KMeansConsolidate}(e_i)$; Flush) by a deferred operator
 184 $\text{Finalize} = \text{KMeansConsolidate}(\{e_i\}_{i=1}^N) \circ \text{Flush}$, executed once after all episodic

Algorithm 1 Soft O2 session soft binding (implementation order)

Require: Query q , dense/associative hits $\mathcal{R}_0$, top- k , attenuation β

- 1: $\mathcal{R} \leftarrow \text{ExpandSiblings}(\mathcal{R}_0; \beta)$ $\triangleright$ raw-score Soft O2
 - 2: $\mathcal{R} \leftarrow \text{ExactBoost/SparseAugment}(q, \mathcal{R})$ $\triangleright$ optional
 - 3: $\mathcal{R} \leftarrow \text{SessionFirstRerank}(\mathcal{R}, k)$
 - 4: fuse each $r \in \mathcal{R}$ with Eq. (1)
 - 5: $\mathcal{R} \leftarrow \text{EnsureSessionCompleteness}(\mathcal{R}; \beta)$ $\triangleright$ final Soft O2 pass
 - 6: **return** top- k of $\mathcal{R}$ by descending fused score
-

185 writes. O3 does not change retrieval semantics; it makes $M \geq 400$ shared-
186 store experiments runnable (≈ 11 min wall-clock for $M=400$; amortized $\approx$
187 0.825 s/turn over ~ 800 turns). Quality–latency curves to $M=6400$ appear
188 in Section 5.6.

189 3.3. Soft O2: session soft binding

190 *Motivation..* Under same-domain interference, a retrieved question turn can
191 score highly while its answer sibling remains outside top- k —episode incom-
192 pleteness compounded by IVFPQ score collapse and context fragmentation.
193 Soft O2 completes episodes without hard score copy.

194 **Definition 3** (Soft binding operator). *Given a scored set $\mathcal{R} = \{(t, s_t)\}$, a*
195 *sibling relation $\mathcal{N}(\cdot)$, and attenuation $\beta \in (0, 1]$, define*

$$\text{SoftBind}(\mathcal{R}; \mathcal{N}, \beta) : \quad s_{t'} \leftarrow \max(s_{t'}, \beta \cdot s_t) \quad \text{for all } t \in \mathcal{R}, t' \in \mathcal{N}(t). \quad (3)$$

196 *Soft O2 is $\text{SoftBind}(\cdot; \text{Sib}, \beta)$ with default $\beta=0.98$. Hard parent hydration is*
197 *the special case $\beta=1$ with forced insertion of all siblings; session-max replaces*
198 *each member score by $\max_{t \in \text{sid}} s_t$.*

199 The released Soft O2 pipeline expands session siblings on raw dense scores,
200 optionally applies exact-match boost and sparse augment, performs session-
201 first reranking, fuses semantic and recency scores (Eq. (1)), then runs a final
202 completeness pass before top- k truncation. Algorithm 1 records that full
203 stage order; β acts on the score field present at each expansion stage in the
204 released memory manager.

205 3.4. Soft O2 attenuation and score competition

206 The attenuation β controls how strongly a retrieved turn promotes its ses-
207 sion siblings. Availability of the injected sibling rises with β , while fidelity of

Table 2: Gap-ratio statistics on LegalEp $M=3000$ stores (exact-channel query proxy; strongest measured non-session distractor). The feasible band $\rho < \beta < 1$ motivates the high- β operating region later confirmed by the AH grid search in Table A.4.

Corpus	ρ_{50}	ρ_{95}	Cover@0.90	Default β
LegalEp-DISC	0.746	0.854	0.981	0.98
LegalEp-Lawyer	0.662	0.815	0.998	0.98

the direct hit falls as $\beta \rightarrow 1$. Table 2 summarizes the empirical gap-ratio distribution $\rho = s_d/s_h$ on LegalEp stores under an exact-channel query proxy: median ρ lies well below one, and a high- β operating point near 0.98 covers most measured distractors while remaining strictly below hard hydration ($\beta=1$). We therefore characterize the availability–ranking trade-off with a three-candidate score-competition model and then choose β by an empirical Answer Hit grid search inside that band, rather than by a closed-form optimum alone. A zero-margin hinge objective on $\beta \in (0, 1]$ is uninformative as a selector: its rank term vanishes for all $\beta < 1$.

Definition 4 (Three-candidate score-competition model). *For a query whose dense retrieval directly hits a gold turn h with fused score $s_h > 0$, let a be the gold answer sibling and let d be the strongest non-session distractor in a wide candidate pool, with scores s_a and s_d . Soft O2 rewrites*

$$s'_h = s_h, \quad s'_a = \max(s_a, \beta s_h), \quad s'_d = s_d, \quad (4)$$

and defines the gap ratio $\rho = s_d/s_h$. Because d is the strongest measured distractor, conditions based on ρ are conservative sufficient conditions for beating that distractor, not exact top- k boundary conditions for every competitor.

The binding analysis is most relevant in the soft-injection regime $s_a \leq \beta s_h$, where $s'_a = \beta s_h$.

Proposition 1 (Feasible attenuation band). *Under Definition 4, assume the soft-injection case $s_a \leq \beta s_h$. Then (i) availability against the measured distractor requires $\beta > \rho$; (ii) rank preservation of the direct hit over the injected sibling requires $\beta < 1$, and more generally $s'_a < s_h$; (iii) when $\rho < 1$, any $\beta \in (\rho, 1)$ satisfies both conditions in the active injection case. If $s_a > s_h$ already, rank preservation depends on the observed sibling score rather than on β alone.*

234 *Proof.* If $s_a \leq \beta s_h$, then $s'_a = \beta s_h$. Availability follows from $\beta s_h > s_d$, i.e.
 235 $\beta > \rho$. Rank preservation requires $s'_a < s_h$, which reduces to $\beta < 1$ when
 236 $s_h > 0$. $\square$

237 *Soft-rank visualization and AH grid search..* As a theoretical diagnostic we
 238 also plot the soft pairwise surrogate

$$\mathcal{L}_{\text{sr}}(\beta; \rho) = -\log \sigma(\tau(\beta - \rho)) - \gamma \log \sigma(\tau(1 - \beta)), \quad (5)$$

239 with logistic σ , temperature τ , and rank weight γ . The curve visualizes the
 240 availability–ranking trade-off under the measured ρ pool and is not itself
 241 the parameter selector. Default $\beta=0.98$ is chosen by an Answer Hit grid
 242 search over $\{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$ on CAIL follow-up and LegalEp ex-
 243 act/paraphrase channels (Table A.4): the high-AH plateau $\{0.90, 0.95, 0.98\}$
 244 matches the theoretically preferred band $\rho < \beta < 1$, while hard hydration ($\beta=1$)
 245 is not better and aggressive attenuation ($\beta=0.5$) collapses binding. That
 246 grid-search table is a selection/sensitivity diagnostic and is not the primary
 247 Soft O2 result grids in Tables 5–A.7, which freeze $\beta=0.98$ under the unified
 248 FlatIP rebuild; absolute AH values are therefore not cross-table comparable.
 249 Advice-recall uses the same selected default without a separate gap-ratio fit.

250 Fair controls include hard parent hydration, session-max aggregation, and
 251 shuffled *sid*. On two-turn LegalEp episodes, hard hydration and session-max
 252 coincide under full-sibling expansion, so tables report a single hard-expansion
 253 baseline. Concatenating a full question–answer episode into one retrieval unit
 254 (BM25-joint or dense joint-episode documents) changes the evidence unit
 255 from turns to documents; we treat that design as an alternative indexing
 256 choice rather than a Soft O2 turn-binding baseline (Section 4).

257 3.5. Semantic consolidation and Soft O2-C

258 Semantic consolidation uses target- k KMeans over turn embeddings, pro-
 259 ducing cluster ids consumed by Soft O2-C. Other agglomerative helpers that
 260 may appear in the repository are unused on the Soft O2-C evaluation path.

261 **Definition 5** (KMeans semantic consolidation). *Given embeddings $\{e_i\}$ and*
 262 *target cluster count k , O3 runs $(\text{labels}, \text{centers}) \leftarrow \text{KMeans}(\{e_i\}; k)$ once at*
 263 *finalize time and assigns each turn t a cluster id cid_t , inducing $\text{Clus}(t)$ in Def-*
 264 *inition 1. Empty clusters are dropped and labels are remapped to a contiguous*
 265 *range.*

Algorithm 2 Gated Hybrid Soft O2 + Soft O2-C

Require: Base hits $\mathcal{R}_0$, budgets (k, B_c) , attenuations (β, β_c)

```
1:  $\mathcal{R} \leftarrow \text{SoftBind}(\mathcal{R}_0; \text{Sib}, \beta)$  ▷ session pathway first
2:  $\mathcal{S}_{\text{cov}} \leftarrow \{sid_t : t \in \text{top-}k(\mathcal{R})\}$ 
3:  $\mathcal{T}_{\text{trig}} \leftarrow \{t \in \mathcal{R}_0\}$  ▷ direct dense hits only
4: for each trigger  $t \in \mathcal{T}_{\text{trig}}$  do
5:    $s_{\text{max}} \leftarrow \max\{s_u : u \in \mathcal{R}, cid_u = cid_t\}$ 
6:   for each  $t' \in \text{Clus}(t)$  with  $sid_{t'} \notin \mathcal{S}_{\text{cov}}$ , up to budget  $B_c$  do
7:      $s_{t'} \leftarrow \max(s_{t'}, \beta_c \cdot s_{\text{max}})$ 
8:   end for
9: end for
10: return top- $k$  of  $\mathcal{R}$ 
```

266 **Definition 6** (Soft O2-C). *Soft O2-C is $\text{SoftBind}(\cdot; \text{Clus}, \beta_c)$ with default*
267 *$\beta_c=0.95 < \beta$, reflecting noisier thematic neighbourhoods than session mem-*
268 *bership. A per-cluster sibling budget B_c caps $|\text{Clus}(t) \cap \text{Injected}|$ so large*
269 *thematic clusters cannot flood top- k .*

270 Gated Hybrid (Algorithm 2) injects only *cross-session* cluster siblings
271 relative to sessions already covered by dense /Soft O2 hits, and triggers
272 cluster binding only on direct dense hits. This prevents thematic confusers
273 from displacing same-session siblings already recovered by Soft O2. Soft O2
274 applies when query and gold share *sid*. Soft O2-C is the cluster-pathway
275 operator when gold evidence lies in another session of the same thematic
276 cluster.

277 4. Corpora and Evaluation Protocol

278 Figure 2 sketches the shared-corpus design used throughout the legal eval-
279 uation. Gold needles and same-domain distractors occupy one store. Each
280 query probes that shared archive under fixed interference conditions. Pri-
281 vately rebuilt haystacks are avoided so that system comparisons isolate the
282 retrieval operator.

283 4.1. Datasets and query channels

284 Multi-turn authenticity is evaluated on CAIL2024 consultation dialogues.
285 Gold needles are authentic preliminary and final dialogues, with conversation
286 turns completed by label turns where applicable. Query channels cover the

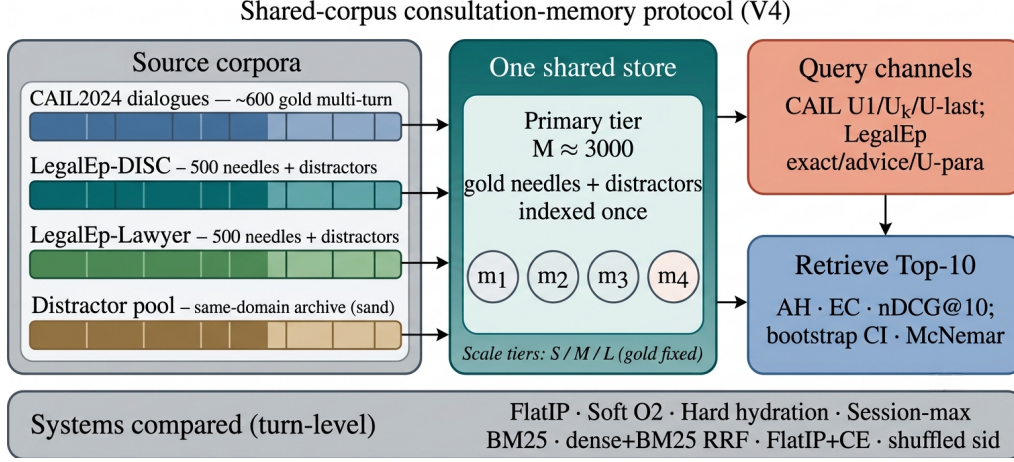


Figure 2: Shared-corpus protocol. Gold needles and same-domain distractors share one store; queries probe multiple channels with turn-level systems and significance tests. Exact replay is diagnostic; paraphrase, follow-up, and advice-recall are primary channels.

287 first user turn ($U1$), a later in-dialogue user turn ($U_k\text{-followup}$), the last user
 288 turn ($U\text{-last}$), and constrained paraphrases ($U\text{-para}$). Evidence defaults to
 289 assistant turns in the target session.

290 LegalEp-DISC and LegalEp-Lawyer rebuild public DISC-Law-SFT and
 291 Lawyer-LLaMA subsets into consultation-episode corpora. Each episode is
 292 a natural (question, answer) pair with session identifier *sid*. Reconstruction
 293 applies length bounds and legal-content filters, excludes exam prompts, re-
 294 moves near-duplicates, separates needles from distractors under a fixed seed,
 295 and retains an audit subsample. After filtering, each LegalEp corpus retains
 296 3500 passed sessions (500 needles, 2500 distractors at the $M=3000$ primary
 297 tier).

298 *Query-channel policy.* Exact replay uses the stored user question and is
 299 reported only as a *diagnostic* upper bound on surface-form recovery. Primary
 300 claims use paraphrase, $U_k\text{-followup}$ / $U\text{-last}$, and advice-recall queries whose
 301 surface form is not identical to the stored question text; paraphrase and
 302 advice-recall templates are generated under a fixed seed and audited on a
 303 subsample. LegalMem-MT reuses CAIL multi-turn gold under the same store
 304 for same-session Soft O2 vs Soft O2-C contrasts and for Mix reconstructions.

305 Archive size is varied with gold needles held fixed. Unless stated otherwise,
 306 Soft O2 transfer tables use the $M \approx 3000$ tier; O1–O2 ablations use $M=400$,

307 $S=300$.

308 4.2. Metrics and baselines

309 Primary metrics are Answer Hit@ k (AH), Episode Completeness@ k (EC),
310 and normalized Discounted Cumulative Gain@ k (nDCG) at $k=10$. AH is
311 binary: whether a gold answer turn appears in the top- k . EC is the mean
312 coverage of gold turns within the target session. Session Hit@ k is reported
313 only for diagnosis. For nDCG we use fixed graded relevance over the full
314 gold session: answer turns $rel=1.0$, other same-session turns $rel=0.5$, all
315 other turns $rel=0.0$. The ideal denominator (IDCG) is computed from all
316 gold-relevant turns in the target session and truncated at k , so every system
317 on a query shares the same IDCG. Answer Hit and Episode Completeness
318 are the main Soft O2 outcomes; nDCG is reported with the fixed gold-session
319 IDCG so that systems remain comparable on each query. Hard expansion
320 can raise nDCG when copied siblings occupy early ranks, so AH and EC are
321 read together with nDCG.

322 The primary hypothesis family is Soft O2 vs FlatIP on the nine CAIL
323 and LegalEp transfer channels; for that family we report McNemar mid- p
324 on paired per-query Answer Hit and Holm-adjusted p -values (Table A.2).
325 Those paired outcomes are taken from the same unified FlatIP rebuild that
326 supplies the Soft O2 grids, so a reader can regenerate both the point estimates
327 and the significance tests from one artifact family. Appendix Soft O2 grids
328 report AH/EC/nDCG point estimates from that rebuild without separate CI
329 columns. Hard-hydration, CE, and Soft O2-C Mix contrasts are secondary.

330 Unless noted, Soft O2 grids in Sections 5.3–5.4 and Appendix Tables A.5–
331 A.7 use one FlatIP rebuild so that AH, EC, nDCG, failure rates, and the
332 Table A.2 paired outcomes on each row come from the same per-query
333 rankings. Consequently, each Δ AH in Table A.2 equals the corresponding
334 Soft O2–FlatIP difference in the main Soft O2 tables (up to ordinary round-
335 ing).

336 Turn-level dense FlatIP (O1) is the base dense retriever. Soft O2 applies
337 soft sibling score inheritance on the episodic pathway; Soft O2-C applies the
338 same rule within KMeans clusters. Hard parent hydration expands a hit
339 to all siblings by copying the trigger score. Lexical controls include BM25
340 over turns and dense+BM25 Reciprocal Rank Fusion (RRF) under the same
341 turn-level evidence unit. BM25-joint and dense joint-episode indexes retrieve
342 pre-concatenated episode documents and therefore answer a coarser retrieval
343 problem; they are discussed as alternative designs in Section 6.3 rather than

344 Soft O2 turn-binding baselines. A cross-encoder (CE) control reranks the
 345 top-30 FlatIP candidates with BAAI/bge-reranker-v2-m3. Shuffled *sid*, a
 346 β sweep, and IVFPQ provide membership, attenuation, and quantization
 347 controls.

348 4.3. Fair Mix protocol for Soft O2-C

349 On fully same-session LegalMem grids Soft O2 already recovers gold an-
 350 swers that share *sid*, leaving Soft O2-C little exclusive headroom and expos-
 351 ing it to thematic confusers. To study off-session recovery we rebuild the
 352 *same* LegalEp and LegalMem archives under a Mix protocol: 70% of gold
 353 episodes are split into query-side S_a and evidence-side S_b with distinct ses-
 354 sion identifiers, while 30% remain intact. Distractors are unchanged. FlatIP,
 355 Soft O2, Soft O2-C, and gated Hybrid share unrestricted dense retrieval on
 356 this store; we report overall AH and stratum AH on cross-session vs same-
 357 session queries.

358 *Cluster co-membership and purity under Mix..* After KMeans consolidation,
 359 each cross-session S_a/S_b pair is assigned a shared cluster identifier. This
 360 construction uses split metadata available only when building the evaluation
 361 store; it does not inject labels at query time. The intent is to test Soft O2-
 362 C when thematic co-membership is present. Cluster purity still matters: a
 363 shared *cid* that also contains many thematically adjacent distractors narrows
 364 Soft O2-C’s effective margin, which is why Hybrid gates cluster injection to
 365 direct dense hits and caps the per-cluster budget B_c . If recovery still fails
 366 under guaranteed S_a/S_b co-membership, the failure is attributable to binding
 367 rather than accidental cluster separation. Natural same-cluster rates without
 368 this construction are lower (Section 6.3). Soft O2’s session membership rule
 369 is unchanged. LegalEp Mix uses advice-recall queries; LegalMem Mix uses
 370 mid-split Uk-followup queries. LongMemEval /LoCoMo numbers from prior
 371 runs on this codebase (QA correctness 70.7% / 68.2%) supply long-horizon
 372 memory context only Wu et al. (2024); Maharana et al. (2024).

373 5. Experiments

374 Unless noted, all dense retrieval grids use the multilingual encoder `nomic-`
 375 `embed-text-v2-moe` Nussbaum et al. (2025) (768-d) as a single shared em-
 376 bedding backend, with $k=10$ and seed 42 on the main $M \approx 3000$ stores. The
 377 encoder is held fixed so Soft O2 vs FlatIP contrasts isolate binding rather

378 than representation changes; sensitivity to alternative Chinese or multilingual
 379 encoders (e.g., `bge-large-zh-v1.5`) is discussed in Section 6.3. A five-seed
 380 check on a smaller shared store ($M=400$) is reported later. Primary contrasts
 381 appear in Figures 3–5; full numeric grids are in the appendix. Each main
 382 table maps to a released JSON artifact in the accompanying repository.

383 5.1. Implementation operators and Soft O2 ablation

384 Table 3 reports the shared-store ablation on the DISC-Law and Lawyer-
 385 LLaMA corpora ($M=400$, $S=300$). IVFPQ baseline Answer Hit is 0.540 on
 386 DISC and 0.397 on Lawyer. O1 alone yields a small gain on DISC (0.567)
 387 and a near-tie on Lawyer (0.397). Exact indexing is therefore a necessary
 388 prerequisite, but it is insufficient alone. O1+Soft O2 raises Answer Hit to
 389 0.780 (DISC) and 0.680 (Lawyer), gains of +0.240 and +0.283 over baseline,
 390 with Episode Completeness 0.888 and 0.840. O3 is enabled for all configura-
 391 tions in this ladder as a construction prerequisite; it does not alter retrieval
 392 scores.

Table 3: O1+Soft O2 ablation on the shared store ($M=400$, $S=300$, $k=10$). EC denotes Episode Completeness (gold-turn coverage); AH denotes Answer Hit. O3 bulk-load is on for all rows. Boldface: best AH and EC within each corpus. Graded nDCG for the $M\approx 3000$ grids appears in Table 6.

Corpus	System	EC@10	AH@10
DISC-Law	IVFPQ baseline	0.770	0.540
	+ O1 FlatIP	0.782	0.567
	+ O1+O2 Soft O2	0.888	0.780
Lawyer-LLaMA	IVFPQ baseline	0.698	0.397
	+ O1 FlatIP	0.698	0.397
	+ O1+O2 Soft O2	0.840	0.680

393 5.2. Failure taxonomy

394 We classify each query into four mutually exclusive modes: *complete*
 395 (question and answer in top- k), *incomplete* (question hit / answer miss),
 396 *answer-only* (answer hit / question miss), and *session miss* (no gold-session
 397 turn in top- k). Table 4 reports these rates for FlatIP, Soft O2, hard hydration,
 398 and shuffled *sid* on primary non-exact channels under the unified FlatIP re-
 399 build. Soft O2 converts many incomplete FlatIP cases into complete retrieval

Table 4: Failure taxonomy at $k=10$ (values in %; rows sum to 100 within rounding). DISC/Lawyer denote LegalEp corpora. Classes: complete (Q+A), incomplete (Q hit / A miss), answer-only, and session miss. Boldface: best complete / lowest incomplete among FlatIP, Soft O2, and shuffled *sid*; hard hydration is unrestricted score copy.

Channel	System	Comp.	Inc.	A-only	Miss
DISC / u-para	FlatIP	68.8	28.8	0.8	1.6
DISC / u-para	Soft O2	88.1	9.9	0.2	1.8
DISC / u-para	Hard	97.2	0.0	0.0	2.8
DISC / u-para	Shuf. O2	61.6	35.8	0.4	2.2
DISC / advice	FlatIP	47.2	17.6	0.0	35.2
DISC / advice	Soft O2	61.0	3.6	0.0	35.4
DISC / advice	Hard	64.8	0.0	0.0	35.2
DISC / advice	Shuf. O2	43.4	21.2	0.0	35.4
Lawyer / u-para	FlatIP	66.9	29.5	2.8	0.8
Lawyer / u-para	Soft O2	94.6	2.8	1.0	1.6
Lawyer / u-para	Hard	97.8	0.0	0.0	2.2
Lawyer / u-para	Shuf. O2	64.9	30.7	2.6	1.8
CAIL / Uk	FlatIP	43.2	56.7	0.0	0.2
CAIL / Uk	Soft O2	93.2	6.7	0.0	0.2
CAIL / Uk	Hard	99.7	0.2	0.0	0.2
CAIL / Uk	Shuf. O2	31.3	68.5	0.0	0.2

while session-miss stays nearly fixed; shuffled *sid* restores high incompleteness. Hard hydration can drive incompleteness near zero by score copy and often leads Soft O2 on complete rate; Soft O2 remains the attenuated binder ($\beta < 1$) rather than a hard session dump (Section 5.4).

5.3. Turn-level Soft O2 controls

Primary Soft O2 claims use a shared turn-level index. Table 5 reports Answer Hit for FlatIP, Soft O2, hard hydration, and shuffled *sid* on the primary Soft O2 transfer channels. Soft O2 beats FlatIP on every non-exact channel; shuffled *sid* collapses toward or below FlatIP. Hard hydration—full sibling score copy—can exceed Soft O2 AH on the same rebuild (Table 5); Soft O2 is best read as an attenuated binder that recovers missing answers without unrestricted score copy (Tables A.5–A.7). BM25-turn is a same-granularity lexical control (Table A.8). Joint episode / document-level indexes change evidence granularity and are scoped as alternative designs rather than Soft O2 turn-binding baselines (Section 6.3).

Table 5: Primary Soft O2 controls (AH@10, $M \approx 3000$, unified FlatIP rebuild). Boldface: best AH within each row among FlatIP / Soft O2 / Hard / Shuffled. Soft O2 is compared primarily to FlatIP; hard hydration is unrestricted score copy; shuffled *sid* tests membership dependence.

Corpus / channel	FlatIP	Soft O2	Hard	Shuffled
CAIL / U1	0.803	0.948	0.995	0.660
CAIL / Uk	0.432	0.932	0.997	0.313
CAIL / U-last	0.393	0.957	0.998	0.255
LegalEp-DISC / u-para	0.696	0.883	0.972	0.620
LegalEp-DISC / advice	0.472	0.610	0.648	0.434
LegalEp-Lawyer / u-para	0.697	0.956	0.978	0.675
LegalEp-Lawyer / advice	0.416	0.654	0.674	0.396

5.4. Soft O2 on session (CAIL and LegalEp)

Figure 3 and Table 5 summarize AH@10 on authentic CAIL multi-turn channels at $M \approx 3000$ under O1 FlatIP. Soft O2 lifts Answer Hit over FlatIP on every channel under the unified rebuild (U1 0.948 vs 0.803; Uk 0.932 vs 0.432; U-last 0.957 vs 0.393). Shuffled *sid* removes the gain, tying the improvement to correct episode membership. Hard expansion can raise AH further by unrestricted score copy (e.g., Uk 0.997 vs Soft O2 0.932). Soft O2 is the attenuated binder: siblings enter under $\beta < 1$ scores, whereas hard hydration copies sibling scores without attenuation. Table 6 shows the Soft O2>FlatIP ordering under graded nDCG with gold-session IDCG, while incompleteness falls sharply (Table 4). Full numeric grids appear in Tables A.5–A.7.

Figure 4 reports LegalEp transfer on the same AH grids. Under the unified rebuild Soft O2 also lifts exact replay (DISC 0.916 vs FlatIP 0.714; Lawyer 0.976 vs 0.678), but we still treat exact as diagnostic relative to paraphrase and advice-recall. Paraphrase yields large Soft O2 gains over FlatIP: 0.883 vs 0.696 on DISC and 0.956 vs 0.697 on Lawyer. Advice-recall similarly rises to 0.608 from FlatIP 0.480 on DISC and to 0.652 from 0.444 on Lawyer. Hard hydration can still edge Soft O2 by unrestricted score copy on these channels; Soft O2 remains the attenuated alternative ($\beta < 1$). Shuffled *sid* collapses toward FlatIP. Graded nDCG mirrors the Soft O2>FlatIP ordering on paraphrase and advice-recall (Table 6). Detailed grids are in Tables A.6–A.7.

Cross-encoder reranking is a strong neural control at the same turn-level evidence unit: FlatIP+CE reranks the top-30 FlatIP candidates with **bge-**

Table 6: Graded nDCG@10 with gold-session IDCG (answer $rel=1.0$, other same-session turns $rel=0.5$) and incompleteness rates on primary non-exact channels. Boldface: Soft O2 better than FlatIP within each channel on nDCG. Soft O2 also exceeds shuffled *sid*.

Corpus	Channel	System	nDCG@10	Incomplete
CAIL	Uk	FlatIP	0.428	56.7%
CAIL	Uk	Soft O2	0.796	6.7%
CAIL	Uk	Shuffled O2	0.349	68.5%
CAIL	U-last	FlatIP	0.400	60.5%
CAIL	U-last	Soft O2	0.796	4.2%
CAIL	U-last	Shuffled O2	0.326	74.3%
LegalEp-DISC	u-para	FlatIP	0.659	28.8%
LegalEp-DISC	u-para	Soft O2	0.742	9.9%
LegalEp-DISC	u-para	Shuffled O2	0.590	35.8%
LegalEp-DISC	advice	FlatIP	0.454	17.6%
LegalEp-DISC	advice	Soft O2	0.518	3.6%
LegalEp-DISC	advice	Shuffled O2	0.411	21.2%
LegalEp-Lawyer	u-para	FlatIP	0.708	29.5%
LegalEp-Lawyer	u-para	Soft O2	0.832	2.8%
LegalEp-Lawyer	u-para	Shuffled O2	0.656	30.7%
LegalEp-Lawyer	advice	FlatIP	0.457	26.2%
LegalEp-Lawyer	advice	Soft O2	0.569	2.0%
LegalEp-Lawyer	advice	Shuffled O2	0.424	27.8%

reranker-v2-m3. Table 7 reports a secondary CE campaign (separate from the unified Soft O2 rebuild): within that campaign FlatIP+CE is strongest on LegalEp exact channels, while Soft O2 leads on CAIL multi-turn channels (Uk 0.698 vs CE 0.375; U-last 0.762 vs 0.342). A structural limit of the reranker explains the split: CE can refine ranking among candidates already retrieved by FlatIP, but it cannot promote an answer sibling that never entered the top-30 pool. Soft O2 recovers such siblings by propagating attenuated scores through session links, which helps precisely when the stored advice turn has weak direct lexical overlap with the query (paraphrase and advice-recall under the unified rebuild; Tables 5 and 6).

5.5. Soft O2-C and gated Hybrid under Mix

When gold answers already share *sid* with the query, Soft O2 is sufficient and cluster binding is unnecessary. Table 8 asks the off-session question under the Mix protocol of Section 4.3: 70% cross-session Sa/Sb gold and 30% intact same-session gold, with FlatIP, Soft O2, Soft O2-C, and gated Hybrid sharing

CAIL2024 multi-turn results ($M \approx 3000$, turn-level systems)

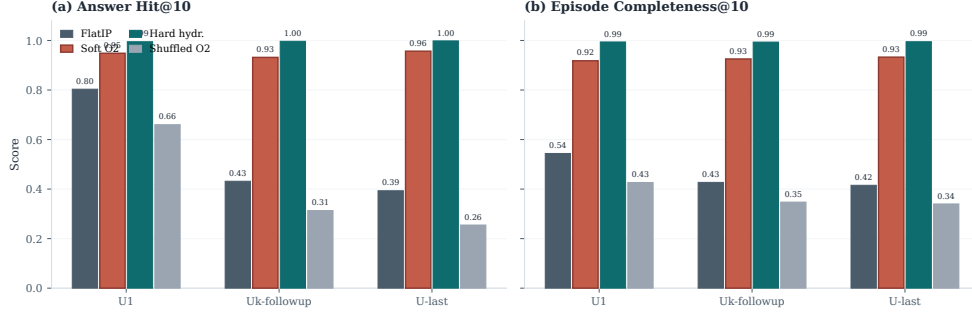


Figure 3: CAIL2024 multi-turn results at tier $M \approx 3000$. Soft O2 improves AH and EC over FlatIP; shuffled *sid* removes the gain; hard expansion can further raise AH/EC by unrestricted score copy.

Table 7: Secondary cross-encoder control (AH@10; CE campaign store). FlatIP+CE reranks the top-30 FlatIP candidates with a multilingual cross-encoder. Soft O2 column is from the same CE campaign for within-table comparison; primary Soft O2 vs FlatIP AH uses the unified rebuild (Tables 5–A.7). Boldface: best AH within each row.

Corpus / channel	FlatIP	FlatIP+CE	Soft O2
CAIL / U1	0.585	0.688	0.788
CAIL / Uk	0.265	0.375	0.698
CAIL / U-last	0.228	0.342	0.762
LegalEp-DISC / exact	0.572	0.722	0.572
LegalEp-Lawyer / exact	0.636	0.708	0.636

unrestricted dense retrieval. Hybrid uses a direct-dense gate: cluster binding fires only on turn ids recorded *before* Soft O2 session expansion.

On LegalEp Mix stores, gated Hybrid is the strongest overall system (DISC AH 0.542, Lawyer 0.536; McNemar mid- $p=0.0005^{**}$ / 0.0022^{**} vs Soft O2) and also leads AH_{cross} . Ungated Soft O2-C raises DISC AH_{cross} to 0.506 from Soft O2 0.460 and Lawyer cross to 0.480 from 0.440, but remains weaker than Hybrid once session binding is composed under the gate. On LegalMem-MT Mix the picture reverses: Soft O2 retains the best overall AH (0.602), while Soft O2-C (0.440) and Hybrid (0.562) trail; FlatIP even leads the cross stratum (0.520). Cluster binding is therefore conditional on corpus and cluster purity—helpful as a gated add-on on LegalEp advice-recall Mix, but not uniformly beneficial on CAIL multi-turn Mix under the same

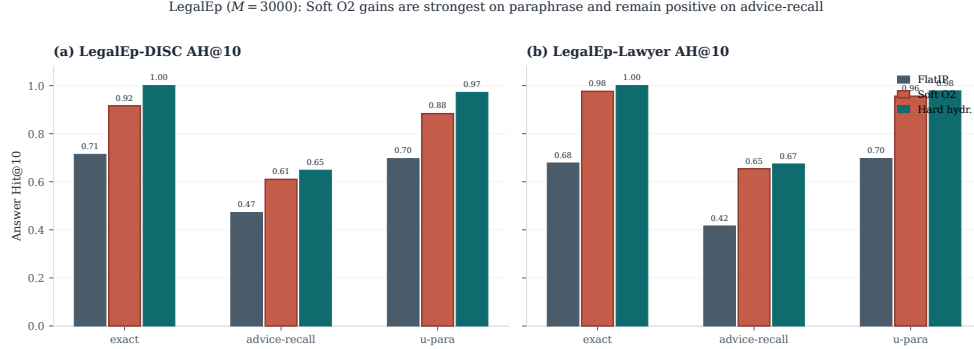


Figure 4: LegalEp AH@10 ($M=3000$, 500 needles). Soft O2 gains are strongest on paraphrase and remain positive on advice-recall; exact transfer is corpus-dependent and treated as diagnostic.

466 protocol. Same-session Soft O2 vs Soft O2-C contrasts remain Soft O2-led
 467 (Appendix Appendix A, Table A.1).

468 Hybrid uses the direct-dense gate. Soft O2-C mid- p vs Soft O2 may favor either side; on
 469 LegalMem-MT the ** marks Soft O2-C significantly *below* Soft O2. Cluster co-membership
 470 of Sa/Sb pairs follows Section 4.3.

471 5.6. Scale behaviour

472 Figure 5 and Table A.3 report quality and warm-query latency on the
 473 LegalEp DISC exact channel ($Q=100$; three repeats; archive sizes 100, 400,
 474 1600, and 6400). Soft O2 retains an AH advantage at each tier. IVFPQ
 475 collapses at $M=1600$ (AH 0.330), which motivates O1, while absolute latency
 476 grows with M . We treat O1 FlatIP as an engineering prerequisite for Soft O2
 477 at these scales and do not claim deployment readiness beyond the measured
 478 curve.

479 *End-to-end QA audit (summary)*.. A dual-protocol QA audit ($N=270$ per
 480 corpus; generator **qwen3:14b**, independent judge **qwen3:32b**) reports judged
 481 answerability under Soft O2 retrieval alone; it is not a paired FlatIP vs
 482 Soft O2 generation experiment. Full protocol and stratified results appear in
 483 Section 6.2. On a smaller shared store ($M=400$, five seeds), Soft O2 AH on
 484 LegalEp-DISC is 0.851 ± 0.008 versus FlatIP 0.698 ± 0.040 .

Table 8: Mix evaluation of Soft O2-C and gated Hybrid with the direct-dense cluster trigger (same store, unrestricted dense; AH@10). Gold is 70% cross-session Sa/Sb + 30% intact same-session. Boldface: best overall AH and best cross-session AH within each corpus. Significance vs Soft O2: * $p<0.05$, ** $p<0.01$.

Corpus	System	AH	AH _{cross}	AH _{same}	mid- p vs Soft O2
LegalEp-DISC Mix	FlatIP	0.494	0.491	0.500	—
	Soft O2	0.492	0.460	0.567	—
	Soft O2-C	0.498	0.506	0.480	0.7275
	Hybrid	0.542	0.526	0.580	0.0005**
LegalEp-Lawyer Mix	FlatIP	0.460	0.451	0.480	—
	Soft O2	0.492	0.440	0.613	—
	Soft O2-C	0.464	0.480	0.427	0.1284
	Hybrid	0.536	0.520	0.573	0.0022**
LegalMem-MT Mix	FlatIP	0.530	0.520	0.553	—
	Soft O2	0.602	0.489	0.867	—
	Soft O2-C	0.440	0.460	0.393	0.0000**
	Hybrid	0.562	0.446	0.833	0.0594

6. Discussion

6.1. Findings and scope

Results show that episode incompleteness is a substantial FlatIP failure mode under same-domain interference, and that Soft O2 converts many incomplete retrievals into complete ones without shrinking session-miss rates (Table 4). O1 restores score fidelity; Soft O2 supplies the main gain through soft sibling inheritance; O3 makes large shared stores tractable. The gap-ratio analysis (Section 3.4) motivates keeping β close to but below unity; the AH grid search then peaks in $\{0.9, 0.95, 0.98\}$ rather than at hard hydration ($\beta=1$). On CAIL under the unified rebuild, Soft O2 raises Answer Hit by large margins over FlatIP (Uk 0.932 vs 0.432; U-last 0.957 vs 0.393) with collapsing shuffled-*sid* controls. Hard hydration can exceed Soft O2 AH/EC by unrestricted score copy. After Holm adjustment on the Soft O2 vs FlatIP family (Table A.2), the non-exact CAIL and LegalEp gains remain significant. Under the unified rebuild, LegalEp exact channels also show Soft O2 AH gains (Tables A.6–A.7); paraphrase and advice-recall remain the main transfer evidence. Graded nDCG agrees with the Soft O2>FlatIP ordering (Table 6). Under Mix, gated Hybrid significantly improves overall AH over Soft O2 on LegalEp-DISC and LegalEp-Lawyer, while Soft O2 remains

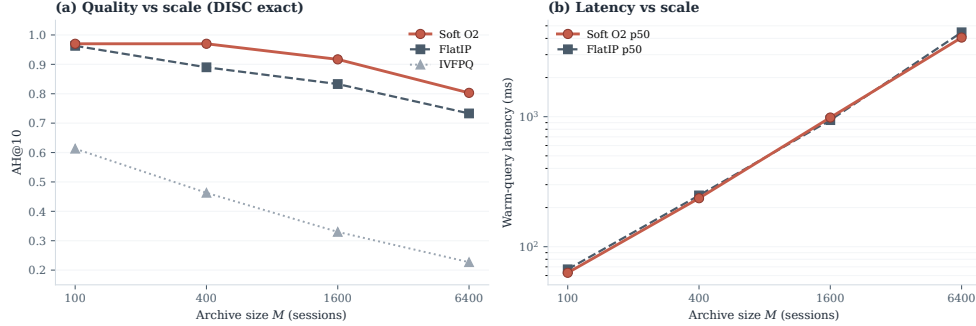


Figure 5: Scale curve on DISC-Law exact. Soft O2 keeps an AH edge over FlatIP as M grows; warm-query latency scales with archive size; IVFPQ quality collapses at moderate M .

strongest on LegalMem-MT (Table 8); Soft O2 stays preferable when gold already shares *sid* (Table A.1). Turn-level BM25, RRF, and CE place Soft O2 among standard IR alternatives Robertson and Zaragoza (2009); Cormack et al. (2009); Nogueira et al. (2019); joint-episode indexes are a coarser design choice outside this protocol.

6.2. RAG application perspective

Consultation-memory retrieval is judged downstream by whether an LLM can answer from recovered evidence rather than confabulate. We audit this end-to-end with a dual-protocol pipeline on Soft O2 retrieval only: generator `qwen3:14b` and independent judge `qwen3:32b` are distinct models. Per corpus we pool $N=270$ judged instances as P1 $n=120$ plus P2 $n=150$. The independent judge assigns a continuous evidence-answerability score in $[0, 1]$ from the retrieved context and the generated reply; Evidence-Answerability Rate (EAR) is the fraction of instances with judge score ≥ 0.7 . Table 9 reports pooled EAR and the mean judge score on a stratified subsample ($n=90$ /corpus; 30 each of exact, paraphrase, and follow-up), labelled “Stratified judge” in the table. In the complete-episode stratum—both question and answer turns in top- k —judged answerability is higher in this single-system audit (pooled EAR 0.867 on DISC and 0.859 on Lawyer; Wilson 95% CI $[0.83, 0.90]$ / $[0.82, 0.89]$). Follow-up queries remain the hardest stratum (independent-judge EAR 0.517 / 0.480), consistent with episodic incompleteness persisting even when surface-form overlap is low.

A blind two-annotator legal audit ($n=60$ /corpus) separates retrieval failures (gold episode absent from top- k) from generation failures given retrieved

evidence. Annotators used a binary answerability label on each instance; agreement is Cohen’s κ computed on those paired labels (0.71 on DISC, 0.68 on Lawyer). Retrieval-failure share is 0.283 / 0.317. Wrong-session contamination rates stay low (0.083 / 0.100), suggesting that complete episode retrieval under Soft O2 is associated with lower rates of a common RAG failure mode in which the model answers from a neighbouring client’s matter. Hallucinated-legal-basis rates (0.117 / 0.133) remain non-zero, indicating that grounding in retrieved episodes does not guarantee statutory correctness—a scope boundary for deployment. AH and EAR measure evidence availability and judged answerability respectively; statutory correctness, citation validity, and professional liability lie outside this evaluation.

Table 9: End-to-end QA audit ($N=270$ per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`). Pooled EAR: fraction with judge score ≥ 0.7 . Stratified judge: mean judge score on $n=90$ /corpus (exact/paraphrase/follow-up). Human κ : Cohen’s κ on binary answerability labels from two blind annotators ($n=60$ /corpus).

Corpus	Pooled EAR	Stratified judge	Human κ
DISC-Law	0.867	0.757	0.71
Lawyer-LLaMA	0.859	0.729	0.68

6.3. Limitations

LegalEp episodes are consultation pairs; multi-turn authenticity remains reserved for CAIL.

Dependency on session segmentation and cluster quality. Soft O2 relies on correct session identifiers (*sid*). In uncured client logs, boundary noise—mid-session topic shifts, multi-intent interruptions, or automated segmentation drift—can corrupt sibling sets. If unrelated turns share a *sid*, attenuated inheritance may propagate false context; if a true advice turn is split into another session, Soft O2 cannot bind it. Soft O2-C likewise depends on KMeans cluster purity. Under the Mix protocol, Sa/Sb pairs receive a shared cluster id so binding can be tested when thematic co-membership is present (Section 4.3); in fully unsupervised consolidation, noisier boundaries shrink the margin over dense retrieval. Deployments on raw consultation streams therefore need reliable upstream session segmentation, and Soft O2-C should be treated as gated rather than always-on.

554 *Joint-episode indexing..* Concatenating full question–answer pairs into
 555 episode documents prevents incompleteness by changing the evidence unit
 556 from turns to merged documents (Section 2). That design lies outside the
 557 Soft O2 turn-binding protocol. A matched comparison under online updates,
 558 latency, and fine-grained advice isolation is left to future work.

559 *Embedding model and other scope limits..* All main grids use `nomic-embed-`
 560 `text-v2-moe`. A Chinese-specialized encoder such as `bge-large-zh-v1.5`
 561 could alter absolute dense ranks and the measured ρ distribution, and might
 562 shrink or widen Soft O2’s relative gain wherever the bi-encoder already places
 563 answer turns inside top- k . We did not re-run the full Soft O2 grids under al-
 564 ternative encoders; the binding operator itself is encoder-agnostic once fused
 565 scores are available. The ρ statistics use an exact-channel query proxy and
 566 the strongest measured distractor; paraphrase and advice-recall gap laws may
 567 shift the feasible band slightly, though the AH sweep keeps a stable high- β
 568 region on those channels. Main $M \approx 3000$ tables use a fixed seed; five-seed vari-
 569 ance is reported only for the smaller $M=400$ store. LLM paraphrases need
 570 human audit. Statutory correctness, citation validity, and professional liabil-
 571 ity remain unevaluated. Because many channels are tested, Holm-adjusted
 572 p -values accompany the Soft O2 vs FlatIP family (Table A.2); Mix Hybrid sig-
 573 nificance is reported only vs Soft O2 within Table 8 and is corpus-conditional.

574 *Latency and deployment scope..* Scale curves cover archive sizes 100, 400,
 575 1600, and 6400 under a fixed hardware stack (Table A.3). At $M=6400$, warm-
 576 query p50 latency exceeds 4 s (4061 ms for Soft O2; 4458 ms for FlatIP)—
 577 unacceptable for millisecond interactive search. BIMS-LEGAL is there-
 578 fore best positioned for high-precision offline or near-real-time consultation-
 579 memory recovery—for example, pre-meeting case review, audit replay, or
 580 batch RAG over a firm’s advice archive—rather than as a sub-millisecond
 581 search engine. Future work will combine approximate nearest-neighbour in-
 582 dexes (e.g., HNSW) with Soft O2 and Soft O2-C binding layered on top
 583 of faithful candidate retrieval, preserving episode completion while reducing
 584 query latency. Behavior at roughly 10^5 sessions is not measured, and we do
 585 not claim asymptotic scalability from O3 wall-clock alone.

586 7. Conclusion

587 BIMS-LEGAL recovers prior advice from Chinese legal consultation
 588 archives under same-domain interference with a dual store: an episodic turn

589 index and a KMeans semantic store. Soft O2 is the main algorithmic con-
 590 tribution: attenuated session binding completes episodes without hard score
 591 copy, while hard hydration remains a stronger unrestricted score-copy up-
 592 per bound on the unified rebuild grids. Soft O2-C extends the same idea
 593 to thematic clusters under Mix reconstructions as a conditional mechanism
 594 test. Attenuation $\beta=0.98$ follows a three-candidate score-competition analy-
 595 sis together with an Answer Hit grid search over $\{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$;
 596 O1 FlatIP and O3 bulk-load are the implementation operators that make
 597 the shared-store ablations feasible. On CAIL multi-turn and LegalEp para-
 598 phrase /advice-recall channels, Soft O2 improves Answer Hit and graded
 599 nDCG over FlatIP and cuts incompleteness; hard hydration can further
 600 raise AH/EC by unrestricted score copy, while shuffled-*sid* controls collapse
 601 Soft O2 gains. Under Mix, gated Hybrid recovers additional LegalEp gold
 602 when cluster co-membership is present, while Soft O2 remains preferable on
 603 LegalMem-MT Mix and on intact same-session gold. An end-to-end Soft O2
 604 QA audit associates complete episode retrieval with judged answerability and
 605 low wrong-session contamination, without claiming paired generation supe-
 606 riority over FlatIP. Turn-level BM25, RRF, and CE place Soft O2 among
 607 standard IR alternatives; joint-episode indexing remains a coarser design
 608 outside this protocol.

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612 *Conflict of interest*

613 The authors declare no competing interests.

614 *Ethics approval*

615 This study uses publicly released research corpora and excludes confiden-
 616 tial client data. The system is a research prototype for consultation-memory
 617 retrieval.

618 *Data availability*

619 Code, processed evaluation corpora, and primary Soft O2 metric artifacts
620 (including paired per-query Answer Hit used for Holm tests) are available at
621 URL withheld for anonymous review. (archived release to accompany camera-
622 ready). Upstream corpora remain under their original licenses: CAIL2024
623 consultation tracks; Hugging Face ShengbinYue/DISC-Law-SFT and Skepsu
624 n/lawyer_llama_data. Processed LegalEp /LegalMem artifacts are derived
625 research releases; redistributors should consult each upstream licence before
626 reuse.

627 *Code availability*

628 The BIMS-LEGAL implementation (O1 FlatIP, Soft O2, O3 bulk-load,
629 Soft O2-C), corpus builders, Mix constructors, and the scripts that regenerate
630 Soft O2 tables, figures, and Holm tests from one metric family ship in the
631 same repository.

632 *Author contributions*

633 Author contributions are withheld for anonymous review.

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748 **Appendix A. Numeric grids and controls**

749 Same-session Soft O2-C contrasts, significance tests, scale curves, and full
 750 Soft O2 numeric grids appear below.

Table A.1: Same-session Soft O2 vs ungated Soft O2-C (AH@10). Soft O2 is preferable when gold already shares *sid* with the query; ungated Soft O2-C can surface thematic confusers. Boldface: better AH within each row. Mix Soft O2-C results appear in Table 8.

Corpus / channel	Soft O2	Soft O2-C
LegalEp-DISC / exact	0.836	0.744
LegalEp-DISC / u-para	0.807	0.730
LegalEp-DISC / advice	0.548	0.490
LegalEp-Lawyer / exact	0.818	0.710
LegalEp-Lawyer / u-para	0.817	0.733
LegalEp-Lawyer / advice	0.546	0.448
LegalMem-MT / U1	0.903	0.712
LegalMem-MT / Uk	0.793	0.385

Table A.2: Primary Soft O2 vs FlatIP family: McNemar mid- p on paired Answer Hit from the unified FlatIP rebuild, with Holm step-down adjustment ($m=9$). Each ΔAH matches Soft O2–FlatIP in the main Soft O2 tables; paired outcomes are released with the paper artifacts.

Corpus / channel	Contrast	ΔAH	mid- p	Holm p
CAIL / Uk	O2 vs FlatIP	+0.500	3.69e-69	2.95e-68
CAIL / U1	O2 vs FlatIP	+0.145	1.15e-16	3.44e-16
CAIL / U-last	O2 vs FlatIP	+0.563	3.19e-85	2.87e-84
LegalEp-DISC / exact	O2 vs FlatIP	+0.202	2.32e-19	9.28e-19
LegalEp-Lawyer / exact	O2 vs FlatIP	+0.298	1.16e-39	8.12e-39
LegalEp-DISC / u-para	O2 vs FlatIP	+0.187	1.28e-16	3.44e-16
LegalEp-Lawyer / u-para	O2 vs FlatIP	+0.259	2.07e-32	1.04e-31
LegalEp-DISC / advice	O2 vs FlatIP	+0.138	1.52e-15	1.52e-15
LegalEp-Lawyer / advice	O2 vs FlatIP	+0.238	7.84e-33	4.70e-32

Table A.3: Scale curve on DISC-Law (mean over 3 repeats). Latency is warm-query p50/p95. Boldface: best AH@10 within each M block; lowest p50/p95 within each M block.

M	Mode	AH@10	Build (s)	p50 (ms)	p95 (ms)
100	FlatIP	0.963	3.0	67	80
100	IVFPQ	0.613	25.2	40	54
100	Soft O2	0.970	3.2	63	69
400	FlatIP	0.890	26.5	248	305
400	IVFPQ	0.463	89.5	171	204
400	Soft O2	0.970	30.9	236	313
1600	FlatIP	0.833	359.8	941	1278
1600	IVFPQ	0.330	251.4	669	751
1600	Soft O2	0.917	393.7	985	1172
6400	FlatIP	0.733	4722.8	4458	4969
6400	IVFPQ	0.227	1072.1	2182	3021
6400	Soft O2	0.803	5245.7	4061	4503

Table A.4: Soft O2 AH@10 grid search over $\beta \in \{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$ used with the gap-ratio analysis to select default $\beta=0.98$. Peak performance lies in $\{0.9, 0.95, 0.98\}$; aggressive attenuation ($\beta=0.5$) weakens binding. This selection/sensitivity table is not the primary Soft O2 result grids; absolute AH values are not comparable to the unified FlatIP-rebuild Tables 5–A.7.

Corpus	0.5	0.7	0.9	0.95	0.98	1.0
CAIL / Uk	0.323	0.537	0.697	0.700	0.698	0.662
LegalEp-Lawyer / exact	0.608	0.654	0.754	0.754	0.752	0.752
LegalEp-Lawyer / u-para	0.631	0.637	0.717	0.745	0.759	0.749
LegalEp-DISC / u-para	0.551	0.551	0.618	0.648	0.648	0.642

Table A.5: CAIL2024 shared-corpus results ($M \approx 3000$). Boldface marks the best AH@10 and best EC@10 within each channel (ties bolded jointly). **: Soft O2 vs FlatIP, McNemar mid- $p < 0.01$. nDCG@10 uses gold-session IDCg.

Channel	System	AH@10	EC@10	nDCG@10
U1	FlatIP	0.803	0.544	0.538
	Soft O2	0.948	0.918	0.767
	Hard hydr.	0.995	0.994	0.859
	Shuffled O2	0.660	0.427	0.393
Uk-followup	FlatIP	0.432	0.427	0.428
	Soft O2	0.932	0.925	0.796
	Hard hydr.	0.997	0.994	0.853
	Shuffled O2	0.313	0.348	0.349
U-last	FlatIP	0.393	0.415	0.400
	Soft O2	0.957	0.933	0.796
	Hard hydr.	0.998	0.995	0.861
	Shuffled O2	0.255	0.340	0.326

751 Soft O2 improves AH/EC over FlatIP; hard hydration uses unrestricted score copy
752 and can lead both.

Table A.6: LegalEp-DISC ($M=3000$, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * $p < 0.05$, ** $p < 0.01$ (Soft O2 vs FlatIP, McNemar mid- p). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCg.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.714	0.857	0.700
	Soft O2	0.916	0.958	0.792
	Hard hydr.	1.000	1.000	0.873
	Shuffled O2	0.642	0.821	0.626
advice-recall	FlatIP	0.472	0.560	0.454
	Soft O2	0.610	0.628	0.518
	Hard hydr.	0.648	0.648	0.563
	Shuffled O2	0.434	0.540	0.411
u-param	FlatIP	0.696	0.836	0.659
	Soft O2	0.883	0.932	0.742
	Hard hydr.	0.972	0.972	0.816
	Shuffled O2	0.620	0.797	0.590

Table A.7: LegalEp-Lawyer ($M=3000$, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * $p<0.05$, ** $p<0.01$ (Soft O2 vs FlatIP). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCg.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.678	0.839	0.710
	Soft O2	0.976	0.988	0.851
	Hard hydr.	1.000	1.000	0.880
	Shuffled O2	0.654	0.827	0.656
advice-recall	FlatIP	0.416	0.546	0.457
	Soft O2	0.654	0.664	0.569
	Hard hydr.	0.674	0.674	0.591
	Shuffled O2	0.396	0.535	0.424
u-param	FlatIP	0.697	0.830	0.708
	Soft O2	0.956	0.965	0.832
	Hard hydr.	0.978	0.978	0.847
	Shuffled O2	0.675	0.815	0.656

Table A.8: Turn-level BM25 and dense+BM25 RRF (AH@10). Boldface: higher AH within each row among reported turn-level cells. BM25-joint / joint-episode document indexes are omitted: they change evidence granularity and were not run as Soft O2 protocol competitors.

Corpus	Channel	AH@10
LegalEp-DISC	exact (BM25-turn)	0.802
LegalEp-DISC	advice-recall (BM25-turn)	0.382
LegalEp-DISC	exact (dense+BM25 RRF)	0.780
LegalEp-DISC	advice (dense+BM25 RRF)	0.394
CAIL	U1 / Uk / U-last (BM25-turn)	0.748 / 0.482 / 0.413
LegalEp-Lawyer	exact / advice (BM25-turn)	0.784 / 0.460